

EDITORIAL

THE SIGNIFICANCE OF SATURATION

An edict of qualitative research is "to collect data until saturation occurs." Yet although we repeatedly read this statement in qualitative methods texts and look for this statement when critiquing in published articles, little thought has been given to the rationale and principles underlying saturation. Thus, as saturation is the key to excellent qualitative work, the purpose of this methodological comment is to explore the significance of saturation.

Saturation is defined as "data adequacy" and operationalized as collecting data until no new information is obtained. Margaret Mead is purported to note that one index of saturation was the boredom that occurred when investigators had "heard it all."

In qualitative research, there are no published guidelines or tests of adequacy for estimating the sample size required to reach saturation equivalent to those formulas used in quantitative research. Rather, in qualitative research, the signals of saturation seem to be determined by investigator proclamation and by evaluating the adequacy and the comprehensiveness of the results.

Qualitative data, although initially appearing diverse and disconnected, in the process of saturation, form patterns or themes and begin to make sense. However, there are no specific guidelines for the *a priori* estimation of the amount of data required in each category or theme to create these patterns. Nor are there rules that guide the process of analysis regarding saturation, except that initially no data are discarded or ignored. Rather—and this is important—in the initial stages of analysis, an investigator must give all data equal consideration in the analytic coding procedures. Frequency of occurrence of any specific incident must be ignored. Saturation involves eliciting all forms or types of occurrences, valuing variation over quantity. The fact that participants may report some things repeatedly and report other aspects of the phenomenon less frequently is, at this stage of the

analysis, inconsequential. It is more important for the researcher to "know it all," than to hear things over and over, forcing a false sense of saturation. However, when the domain has been fully sampled—when all data have been collected—then replication of data occurs and, with this replication, Mead's boredom, the signal of saturation. I repeat: The *quantity* of data in a category is not theoretically important to the process of saturation. Richness of data is derived from detailed description, not the number of times something is stated. Frequency counts are out.

This principle is crucial to the understanding of saturation. Further, it is this process that is most confusing to new investigators, because in quantitative methods the significance of numbers is carefully taught, and statistical significance is based on frequencies, averages, and the distribution of data. *Frequency* is central to the analysis, and if a particular instance is too abhorrent, it may even be deleted from the data set as an "outlier" or an error. On the other hand, in qualitative analysis, the converse is true. It is often the infrequent gem that puts other data into perspective, that becomes the central key to understanding the data and for developing the model. It is the *implicit* that is interesting.

Qualitative sampling, based on variation determined by the developing analysis, has two aspects that are scary. The first is, how does the researcher recognize when the results are complete? The second is, how does the researcher know when enough data are enough? Both of these questions are based on the careful—and theoretically justified—delineation of the sample. The tighter and more restrictive the sample and the narrower and more clearly delineated the domain, the faster saturation will be achieved. In reality, studies begin broadly and focus naturally as the researcher homes in on some aspect that is particularly interesting, and the process of inquiry narrows to a particular topic. Researchers cease data collection when they have enough data to build a comprehensive and convincing theory. That is, saturation occurs.

These links or connections between categories may not be frequently occurring in data or may not be immediately apparent to the researcher. For this reason, constructing theory in qualitative research is a highly creative act. It is why qualitative research makes a contribution to knowledge, and why it requires researchers to have intuitive skills and an encyclopedic knowledge of social science theory. Similarly, in quantitative research, two concepts may be actually associated or correlated, but the theoretical rationale for such significance

may be obscure or obtuse, or, if not immediately apparent, the correlation may be ignored or discounted as "spurious." The researcher is always in the driver's seat.

Principles of Saturation in Qualitative Research

1. Select a cohesive sample. The greater the cohesiveness of the sample, the faster saturation will be obtained, but the less generalizability of the project. This includes using a culturally cohesive sample and a sample that shares (with least variation) the characteristics that address the research topic.
2. Saturation will be achieved most quickly if theoretical sampling is used. Snowball, or a convenience sample, will result in saturation being achieved more slowly. With a random sample, saturation may never be achieved because the sample may be theoretically inappropriate, or poor informants, whose stories replicate rather than provide new information, may be randomly selected.
3. Sample all variations appearing within the data until each "negative case" perspective is saturated. When constructing a theory, locate every possible "hypothetical" negative case, and give these data equal attention as the mainstream storyline.
4. Saturated data are rich, full, and complete. The resulting theory makes sense and does not have gaps.
5. The more complete the saturation, the easier it is to develop a comprehensive theoretical model.

The Ramifications of the Lack of Saturation

Failure to achieve saturation not only severely impedes the quality of the research, but it also means that the researcher's task of theory development is more frustrating and more difficult. If one does not have the blocks with which to build, one cannot build. More seriously, not following the principles of theoretical sampling means that the phenomenon has not been fully explored. In short, the results are invalid.

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